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Day 9

CSS

Transitions, Transforms & Animations

Introduction

In modern web design, CSS offers powerful tools to bring websites to life.A few of these tools are transformations,transitions and animations.

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Transform

Property / Function	Description	Values / Examples
`transform`	Applies 2D or 3D transformation	`translate()`, `rotate()`, `scale()`, `skew()`
`translate(x, y)`	Moves element on X and/or Y axis	`translate(50px, 100px)`
`translateX(x)` / `translateY(y)`	Move only along one axis	`translateX(20px)`
`rotate(angle)`	Rotates element clockwise	`rotate(45deg)`
`scale(x, y)`	Resizes element	`scale(1.5)` or `scale(1.2, 0.8)`
`scaleX(x)` / `scaleY(y)`	Resize one direction	`scaleY(2)`
`skew(x-angle, y-angle)`	Slants element	`skew(10deg, 15deg)`
`matrix(...)`	Applies all transforms as a matrix	`matrix(a, b, c, d, e, f)`
`perspective()`	Sets 3D perspective	`perspective(500px)`
`transform-origin`	Point of transformation center	`center`, `top left`, `50% 50%`, etc.

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Transition

⏳ CSS Transitions

Property	Description	Values / Examples
`transition`	Shorthand for all transition properties	`all 0.3s ease-in-out`
`transition-property`	CSS property to animate	`background-color`, `transform`, etc.
`transition-duration`	Time of animation	`0.5s`, `2s`
`transition-delay`	Wait before start	`0s`, `1s`
`transition-timing-function`	Acceleration curve	`ease`, `linear`, `ease-in`, `ease-out`, `ease-in-out`, `cubic-bezier()`

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Animations

🎞️ CSS Animations (`animation` shorthand)

Property	Description	Values / Examples
`animation`	Shorthand for all animation properties	`slide 2s linear infinite`
`animation-name`	The `@keyframes` name	`slide`, `spin`, etc.
`animation-duration`	Time it takes to complete one cycle	`2s`, `5s`
`animation-timing-function`	Speed curve	`ease`, `linear`, `ease-in-out`, etc.
`animation-delay`	Delay before animation starts	`0s`, `2s`
`animation-iteration-count`	Number of times it runs	`infinite`, `1`, `3`, etc.
`animation-direction`	Direction it runs	`normal`, `reverse`, `alternate`, `alternate-reverse`
`animation-fill-mode`	What styles to apply before/after animation	`none`, `forwards`, `backwards`, `both`
`animation-play-state`	Pause/play state of animation	`running`, `paused`

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Key Frames

🗂️ @keyframes

Syntax Element	Description	Example
`@keyframes name`	Defines animation named `name`	`@keyframes bounce { ... }`
`0%`, `100%`	Start and end points	`0% { transform: scale(1); }`
`from`, `to`	Shorthand for 0% and 100%	`from { ... } to { ... }`
Intermediate %	Add more steps in timeline	`50% { transform: scale(1.5); }`
Multiple keyframes	Can define several for complex animations	`0%, 100% { ... }, 50% { ... }`

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HTML

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Ferris Wheel</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
<div class="ferris-container">
  <div class="wheel">
    <div class="cabin" style="--i: 0;"><span>Seat 1</span></div>
    <div class="cabin" style="--i: 1;"><span>Seat 2</span></div>
    <div class="cabin" style="--i: 2;"><span>Seat 3</span></div>
    <div class="cabin" style="--i: 3;"><span>Seat 4</span></div>
    <div class="cabin" style="--i: 4;"><span>Seat 5</span></div>
    <div class="cabin" style="--i: 5;"><span>Seat 6</span></div>
  </div>
</div>



</body>
</html>
```

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CSS

```
/* === GLOBAL STYLES === */
body {
  margin: 0;
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  font-family: sans-serif;
  background: linear-gradient(to top, #dbeafe, #fff);
}

/* === FERRIS WHEEL CONTAINER === */
.ferris-container {
  position: relative;
  width: 300px;
  height: 300px;
}

/* === WHEEL BASE === */
.wheel {
  width: 100%;
  height: 100%;
  border: 8px solid #6e44d9;
  border-radius: 50%;
  display: flex;
  justify-content: center;
  align-items: center;
  position: relative;
  transition: transform 10s linear;
}

/* Start rotating wheel on hover */
.ferris-container:hover .wheel {
  animation: rotateWheel 8s linear infinite;
}

/* === CABINS === */
.cabin {
  position: absolute;
  width: 60px;
  height: 60px;
  border-radius: 50%;
  background-color: #6e44d9; /* default static color */
  color: white;
  font-size: 12px;
  font-weight: bold;
  text-align: center;
```

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CSS

```
/* === CABINS === */
.cabin {
  position: absolute;
  width: 60px;
  height: 60px;
  border-radius: 50%;
  background-color: #6e44d9; /* default static color */
  color: white;
  font-size: 12px;
  font-weight: bold;
  text-align: center;
  text-shadow: 1px 1px 2px #000;
  display: flex;
  justify-content: center;
  align-items: center;
  transform: rotate(calc(var(--i) * 60deg)) translateY(-130px);
}

/* Animate color shift only on hover */
.ferris-container:hover .cabin {
  animation: colorShift 4s infinite alternate;
}

/* Prevent span from catching mouse events */
.cabin span {
  pointer-events: none;
}

/* === ANIMATIONS === */
@keyframes rotateWheel {
  from { transform: rotate(0deg); }
  to   { transform: rotate(360deg); }
}

@keyframes colorShift {
  0%   { background: rgb(255, 133, 133); }
  25%  { background: rgb(255, 232, 158); }
  50%  { background: rgb(134, 134, 255); }
  75%  { background: rgb(145, 255, 189); }
  100% { background: rgb(126, 182, 255); }
}
```